

Independent Study

Deliverable 6 | May 10th, 2026

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This Week

- Making sure LTRAG works
- Literature review

LTRAG Setup Ready

I finished setting up LTRAG. I set up runner.py, which you can find in each framework (FOLIO and LSAT) of myLTRAG found in [aiea-rami](#). I can now run custom tests on both datasets. Important customizations I set up include customizing parallelism and how many examples of each framework you can test:

1. You can choose N_SAMPLES: How many samples of each framework
2. You can choose N_PROCESSES: how many processes/tests run at the same time. For example, testing N FOLIO samples at the same time

There is also a well-formatted output. You can see the path to my results in the “references” section below.

Process Control

This is a key feature LTRAG taught me (process pool). LTRAG triggers n processes at the same time. Each process tests one of the examples up to k samples. The process pool has been the quickest way for me to carry out tests of this nature. Usually the longest part is waiting for the API call to OpenAI. This way we send n API requests at the same time. This process pool has completely changed the way I approach LLM research.

Literature Review

I found a highly similar paper to my research project, [Large Language Models Imitate Logical Reasoning, but at what Cost?](#) Their hypotheses are very similar. In the end, they show that using a small language model with neurosymbolic reasoning is more cost effective and its performance is on par with modern-day LRMs. A few comments are

1. They don't closely follow the scientific method because they test many different models over only one data set (PrOntoQA) that they modified, which can introduce bias.
2. Their study lacks a quantifiable measure of reasoning and its effect on cost and accuracy with and without a neurosymbolic pipeline.

Next Steps

I am waiting on Leilani's input on finding such a similar paper. Until then, I will do a literature review marathon next week because I have not read enough papers.

References

AI Use

I used Claude to

1. Help me format structured printing in stdout
2. Help me understand the pre-existing process pool in more depth

All changes were made by the author

LTRAG working tests stdout on 4o-mini:

Please find initial.txt in [aiea-rami](#)/myLTRAG/initial.txt

